Chapter 7

The Yoneda Lemma

7.1 Hom-sets

Definition 7.1.1 (Hom-sets)**.** For objects A and B in a category $\mathcal{C}$, a *hom-set* $\mathcal{C}(A, B)$ is a set of morphisms with domain A and codomain B.

Remark 7.1.2. Some authors write $\mathcal{C}(A, B)$ as $hom_{\mathcal{C}}(A, B)$ or $Hom_{\mathcal{C}}(A, B)$.

Definition 7.1.3 (Small Categories)**.** A category $\mathcal{C}$ is *small* if its morphisms can be indexed by a set. $\mathcal{C}$ is *locally small* if $\mathcal{C}(A, B)$ can be indexed by a set for any pair of objects A and B.

Remark 7.1.4. Most categories that we use are locally small. However, functor categories are not always so.

Theorem 7.1.5. If $\mathcal{C}$ is small and $\mathcal{D}$ is locally small, then $\mathcal{D}^{\mathcal{C}}$ is locally small.

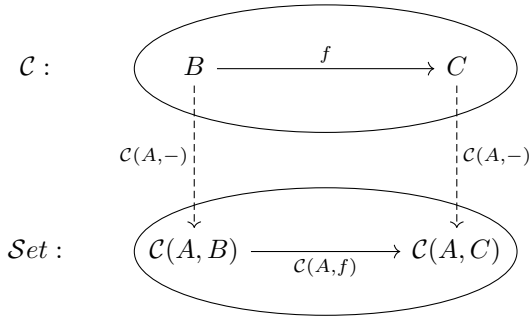
Proof. For a natural transformation $\alpha : \boldsymbol{F} \dot{\longrightarrow} \boldsymbol{G}$ between two functors $\boldsymbol{F}, \boldsymbol{G} : \mathcal{C} \longrightarrow \mathcal{D}$, we have to specify a morphism $\alpha_C : \boldsymbol{F}C \longrightarrow \boldsymbol{G}C$ in $\mathcal{D}$ for each object

C of $\mathcal{C}$. Thus there is a candidate for such a natural transformation α for every function

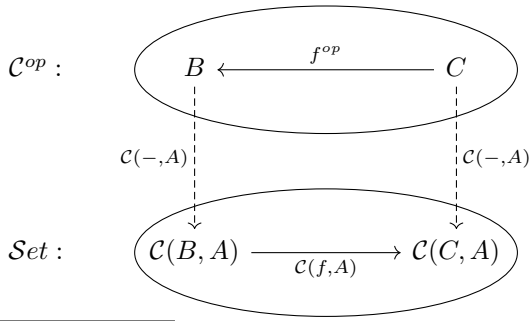
$$ob\mathcal{C} \longrightarrow \bigcup \{\mathcal{D}(\boldsymbol{F}C, \boldsymbol{G}C) \mid C \in ob\mathcal{C}\}$$

and there is certainly a set of such functions. [1] □

Remark 7.1.6. If $\mathcal{C}$ is locally small, a hom-set $\mathcal{C}(A, B)$ is an object in $\mathcal{S}et$. Then we can define a covariant functor $\mathcal{C}(A, -) : \mathcal{C} \longrightarrow \mathcal{S}et$, which assigns an object B in $\mathcal{C}$ to $\mathcal{C}(A, B)$ in $\mathcal{S}et$ and a morphism $f : B \longrightarrow C$ in $\mathcal{C}$ to $\mathcal{C}(A, f) : \mathcal{C}(A, B) \longrightarrow \mathcal{C}(A, C)$ that maps a morphism $g : A \longrightarrow B$ in $\mathcal{C}(A, B)$ to $f \circ g$ in $\mathcal{C}(A, C)$.



Remark 7.1.7. Dually, we can define a contravariant functor $\mathcal{C}(-, A) : \mathcal{C}^{op} \longrightarrow \mathcal{S}et$, which assigns an object B in $\mathcal{C}^{op}$ to $\mathcal{C}(B, A)$ in $\mathcal{S}et$ and a morphism $f^{op} : C \longrightarrow B$ in $\mathcal{C}^{op}$ to $\mathcal{C}(f, A) : \mathcal{C}(B, A) \longrightarrow \mathcal{C}(C, A)$ that maps a morphism $g^{op} : B \longrightarrow A$ in $\mathcal{C}(B, A)$ to $f^{op} \circ g^{op}$ in $\mathcal{C}(C, A)$.



[1] Crole (1993, p.61)

Remark 7.1.8. Moreover, we can define a functor $\mathcal{C}(-,-) : \mathcal{C}^{op} \times \mathcal{C} \longrightarrow \mathcal{S}et$, which assigns a pair of objects A and B to an object $\mathcal{C}(A,B)$ in $\mathcal{S}et$ and a pair of morphisms $f : A \longrightarrow A'$ and $g : B \longrightarrow B'$ to a morphism $\mathcal{C}(f,g)$ that maps a morphism h in $\mathcal{C}(A,B)$ to $h \circ f \circ g$:.

Definition 7.1.9 (Yoneda functor)**.** Since functor categories are cartesian closed, *Yoneda functor* is obtained though transposing the functor $\mathcal{C}(-,-) : \mathcal{C}^{op} \times \mathcal{C} \longrightarrow \mathcal{S}et$.

$$Y : \mathcal{C}^{op} \longrightarrow \mathcal{S}et^{\mathcal{C}}$$

There is also its dual:

$$Y' : \mathcal{C} \longrightarrow \mathcal{S}et^{\mathcal{C}^{op}}$$

7.2 Yoneda Lemma

Theorem 7.2.1 (Yoneda)**.** Let $\mathcal{C}$ be a small category. If $\boldsymbol{K} : \mathcal{C} \longrightarrow \mathcal{S}et$ is a functor and A an object in $\mathcal{C}$, there is a isomorphism:

$$\mathcal{S}et^{\mathcal{C}}(\mathcal{C}(A,-), \boldsymbol{K}) \quad \cong \quad \boldsymbol{K}A$$

which sends each natural transformation $\alpha : \mathcal{C}(A,-) \xrightarrow{\cdot} \boldsymbol{K}$ to $\alpha_A id_A$, and sends each element $a \in \boldsymbol{K}A$ to a natural transformation β whose component β_B sends $f \in \mathcal{C}(A,B)$ to $(\boldsymbol{K}f)a$.[2]

Remark 7.2.2. Recall that $\mathcal{S}et^{\mathcal{C}}$ is a category of functors from $\mathcal{C}$ to $\mathcal{S}et$. Thus $\mathcal{S}et^{\mathcal{C}}(\mathcal{C}(A,-), \boldsymbol{K})$ is a collection of natural transformations from $\mathcal{C}(A,-)$ to $\boldsymbol{K}$. By Theorem 7.1.5, this collection is a set.

Proof. Any element $a \in \boldsymbol{K}A$ is sent to a natural transformation $\beta \in \mathcal{S}et^{\mathcal{C}}(\mathcal{C}(A,-), \boldsymbol{K})$ and then sent back to $\beta_A id_A \in \boldsymbol{K}A$, but this equals to a as shown below:

$$\begin{aligned} \beta_A id_A &= (\boldsymbol{K}id_A)(a) \\ &= id_{\boldsymbol{K}A}(a) \\ &= a \end{aligned}$$

[2] MacLane (1997, p.61:(4))

Conversely, any natural transformation $\alpha \in \mathcal{S}et^{\mathcal{C}}(\mathcal{C}(A,-),\boldsymbol{K})$ is sent to $\alpha_A id_A$ and then sent back to a natural transformation $\beta \in \mathcal{S}et^{\mathcal{C}}(\mathcal{C}(A,-),\boldsymbol{K})$, but this equals to α since each component β_B sends $f : A \longrightarrow B$ to:

$$\begin{array}{rcll}(\boldsymbol{K}f)(\alpha_A id_A) & = & (\boldsymbol{K}f \circ \alpha_A)(id_A) & \\ & = & (\alpha_B \circ \mathcal{C}(A,f))(id_A) & \text{(By naturality of } \alpha) \\ & = & \alpha_B(\mathcal{C}(A,f)(id_A)) & \\ & = & \alpha_B(f \circ id_A) & \text{(By definition of } \mathcal{C}(A,f)) \\ & = & \alpha_B(f) & \end{array}$$ □

There is another isomorphism for the dual Yoneda functor.

Corollary 7.2.3. Let $\mathcal{C}$ be a small category. If $\boldsymbol{K'} : \mathcal{C}^{op} \longrightarrow \mathcal{S}et$ is a functor and A an object in $\mathcal{C}^{op}$, there is a isomorphism:

$$\mathcal{S}et^{\mathcal{C}^{op}}(\mathcal{C}(-,A),\boldsymbol{K'}) \quad \cong \quad \boldsymbol{K'}A$$

which sends each natural transformation $\alpha : \mathcal{C}(-,A) \xrightarrow{\cdot} \boldsymbol{K'}$ to $\alpha_A id_A$, and sends each element $a \in \boldsymbol{K'}A$ to a natural transformation β whose component β_B sends $f \in \mathcal{C}(B,A)$ to $(\boldsymbol{K'}f)a$.

Exercise 7.2.4. Prove the above corollary.

Theorem 7.2.5. Yoneda functor and its dual are full and faithful.

Proof. Apply $\mathcal{C}(B,-)$ to $\boldsymbol{K}$ of the Yoneda lemma. □

7.3 Representable Functor

Definition 7.3.1 (Representation)**.** Let $\mathcal{C}$ be a small category. A *representation* of a functor $\boldsymbol{K} : \mathcal{C} \longrightarrow \mathcal{S}et$ is a pair $\langle R, \psi \rangle$, with R an object of $\mathcal{C}$ and

$$\psi : \quad \mathcal{C}(R,-) \quad \cong \quad \boldsymbol{K}$$

a natural transformation. The object R is called the *representing* object. The functor $\boldsymbol{K}$ is said to be *representable* when such a representation exists.[3]

[3] MacLane (1997, p.60:(3))

Theorem 7.3.2. Every universal arrow is a representable functor, and vise versa.

Proof. (to be written) □

Definition 7.3.3 (Representables)**.**

7.4 End and coEnd

Theorem 7.4.1 (Yoneda Lemma as ends)**.**

$$\int_{c' \in \mathcal{C}} Set(\mathcal{C}(c', c), \boldsymbol{K}c') \quad \cong \quad \boldsymbol{K}c$$

7.5 coYoneda Lemma

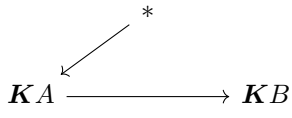
The following statement is called the coYoneda lemma by MacLane which is attributed to Kan.

Theorem 7.5.1 (coYoneda)**.** Suppose that $\mathcal{C}$ is a category, $\boldsymbol{K} : \mathcal{C} \longrightarrow \mathcal{S}et$ is a functor:

$$\mathcal{S}et^{\mathcal{C}}(\boldsymbol{K}, C(A, -)) \quad \cong \quad \mathcal{C}^{(* \downarrow \boldsymbol{K})}(\Delta A, \pi)$$

Remark 7.5.2. $(* \downarrow \boldsymbol{K})$ is a subcategory of the comma category $(\mathcal{S}et \downarrow \boldsymbol{K})$ whose object (f, X) consists of a morphism $f : * \longrightarrow \boldsymbol{F}A$ for some $A \in \mathcal{C}$.

Recall that ΔA is a constant diagram defined in Definition 5.2.2.



[3] MacLane (1997, p.62:Ex)

Theere is an alternative statement for coYoneda:

Theorem 7.5.3 (coYoneda (alternative))**.** Every presheaf $\boldsymbol{K}$ is a colimit of representables, in that:

$$\boldsymbol{K} \quad \cong \quad \int^{c' \in \mathcal{C}} \mathcal{C}(c, c') \otimes \boldsymbol{K} c'$$

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